MIL-D-45471 (Ord)
28 March 1960

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, T76
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of electric detonator designated as Detonator, Electric, T76.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-L-201 - Lead, Sheet

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-14077 - Polytetrafluoroethylene (Teflon) Molding and Extruding Material

FSC 1375

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

8800784 - Ammunition Container, Metal for Detonators, Electric
8826685 - Electric Detonator T76, Assembly
8798698 - Box, Set-Up, Packing, Ammunition for Detonators, Electric
8798699 - Box, Packing, Ammunition for Detonators, Electric

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
EL-8792233 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Detonators.-The detonators shall comply with all requirements specified on Drawing (dwg.) 8826685, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide.-The moisture content of the lead azide at the loading station at the time of loading the detonators shall not exceed 0.50 percent.

3.4 Moisture content of PETN.-The moisture content of the PETN at the loading station at the time of loading the detonators shall not exceed 0.50 percent.

3.5 Functioning at 70 degrees Fahrenheit ($^{\circ}F.$) $\pm$ 10$^{\circ}F.$

3.5.1 Waterproofness.-The detonator shall be waterproof as evidenced by normal functioning after immersion in water for not less than 48 hours at a depth of 2 to 3 inches at a temperature of 20 degrees centigrade ($^{\circ}C.$) $\pm$ 1$^{\circ}C.$

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3.5.2 Time.-The detonator shall fire on the first attempt within 50 microseconds when energy from a .004 microfarad capacitor, maximum (max.) charged to 100 volts max. is applied.

3.5.3 Output.-The detonator shall fire on the first attempt when energy from a .004 microfarad capacitor max., charged to 100 volts max. is applied and shall cause perforation of the lead disc to not less than a minimum (min.) diameter of 3/16 inch.

3.5.4 Duds.-The detonator shall fire on the first attempt when energy from a .004 microfarad capacitor max., charged to 100 volts max., is applied.

3.6 Workmanship.-All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The quality assurance provisions of this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. Inspection shall be performed in accordance with this specification, Specification MIL-G-2550 and other specifications referenced in any of the contractual documents.

4.1.1 Contractor inspection.-Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply.

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The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.3 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of the agent's authority to bind his principal.

4.1.4 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with.

4.1.4.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.4.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.1.4.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. The avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of detonators produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting performance, safety or fit, as determined by the Government, shall not be cause for changing the lot number. Each lot shall contain:

a. metal parts or metal parts assemblies of one interfix lot number from one manufacturere.

b. lead azide from not more than one lot from one manufacturer.

c c. PETN from not more than one lot from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent. for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 42.4.

4.2.2.1 Bridge and plug assembly (prior to application of spotting charge) (see Dwg. 8826687 covering a detail of dwg. 8826685).

	Categories and defects	Method of inspection
Major	AQL 0.25 percent	
1.	Bridge missing	Visual
2.	Lacquer missing from body	Visual

4.2.2.2 Bridge and plug assembly (see dwg. 8826687 covering a detail of dwg. 8826685).

	Categories and defects	Method of inspection
Major	AQL 0.25 percent	
1.	Spotting charge missing or cracked	Visual
2.	Lacquer missing from body	Visual
Minor	AQL 0.40 percent	
41.	Evidence of poor workmanship (see 3.6)	Visual

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4.2.2.3 Assembly (see dwg. 8826685).

	Categories and defects	Method of inspection
Major	AQL 0.25 percent	
1.	Total length, max	Gage
2.	Outside diameter	Gage
Minor	AQL 1.50 percent	
41.	Crimp not full 360 degrees	Visual
42.	Contact surface of pin not clean	Visual
43.	Varnish seal missing from cup crimp	Visual
44.	Evidence of poor workmanship (see 3.6)	Visual
45.	Length of cup, min	Gage

4.2.2.4 Sealed metal container (see dwg. 8800784)

	Categories and defects	Method of inspection
Major	AQL 0.25 percent	
1.	Contents exposed	Visual
2.	Sealing strip incomplete or badly wrinkled	Visual
Minor	AQL 1.00 percent	
41.	Contents move when shaken	Manual
42.	Marking, misleading or unidentifiable	Visual
43.	Label missing (when applicable)	Visual

4.2.2.5 Box, set-up, packing (see dwg. 8798698).

	Categories and defects	Method of inspection
Major	AQL 0.25 percent	
1.	Waterproof seal broken by puncture, cut or tear	Visual
2.	Sealing strip incomplete or badly wrinkled	Visual
Minor	AQL 1.00 percent	
41.	Contents move when shaken	Manual
42.	Marking misleading or unidentifiable	Visual
43.	Label missing (when applicable)	Visual

4.2.2.6 Sealed wood packing box (see dwg. 8798699)

	Categories and defects	Method of inspection
Major	AQL 0.40 percent	
1.	Strapping missing broken or loose	Visual-Manual
2.	Contents exposed	Visual
3.	Board broken or split	Visual
4.	AIC symbol missing or unidentifiable	Visual
Minor	AQL 1.50 percent	
41.	Strapping improperly assembled	Visual-Manual
42.	Car seal missing, unsealed or improperly positioned	Visual-Manual
43.	Contents move when shaken	Manual
44.	Marking misleading or unidentifiable	Visual

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4.2.3 Testing.

4.2.3.1 Resistance between pin and cup (see dwg. 8826685) major defect.- This test shall be performed 100 percent. Any assembly which fails to meet the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.3 using test equipment as specified in 4.2.4.

4.2.3.2 Moisture content of lead azide (see 3.3) and moisture content of PETN (see 3.4) major defects.-A sample of six grains of lead azide and 15 grams of PETN shall be selected, prior to loading, from that quantity used to produce a lot of detonators for determination of moisture content. The determinations shall be performed as specified in 4.3.1 and 4.3.2 respectively.

4.2.3.3 Functioning (see 3.5) Major defect.-One hundred and eighty five detonators shall be selected from each lot for this test.

4.2.3.3.1 Time and duds.-Thirty-five of the 185 detonators selected in 4.2.3.3 shall be used for this test. If the sample mean functioning time ($\bar{x}$) plus the product of 2.27 times the standard deviation (s) exceeds the time specified, the lot shall be rejected. The standard deviation shall be computed from the formula in paragraph 6.2 or equivalent. Duds shall be recorded, but shall not enter into the calculation for standard deviation. All duds are to be replaced by detonators from the 185 selected in 4.2.3.3. This test shall be performed as specified in 4.3.4.2 using test equipment as specified in 4.2.4.

4.2.3.3.2 Output and duds.-Fifty of the 185 detonators, selected in 4.2.3.3, shall be used for this test. If two or more detonators fail to comply with the output requirement, the lot shall be rejected. If only one defective is found, the balance of the detonators selected in 4.2.3.3 shall be tested. If the combined number of defectives in both the first and second samples is two or more, the lot shall be rejected. This test shall be performed as specified in 4.3.4.3 using test equipment as specified in 4.2.4. Duds shall be recorded and replaced by detonators from the 185 selected in 4.2.3.3.

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4.2.3.3.3 Duds.-If no retest is required in the output phase, the balance of the 185 detonators selected in 4.2.3.3 shall be subjected to the test for duds. If three duds are found in the 185 detonators selected in 4.2.3.3 the lot shall be rejected without retest. If two duds are found, a second sample of 370 detonators shall be tested. If the combined number of defectives in both the first and second samples is three or more, the lot shall be rejected. This test shall be performed as specified in 4.3.4.4 using test equipment as specified in 4.2.4.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.2.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment No. 8792233, or approved equivalents thereto. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authroized Government personnel.

4.3 Test methods and procedures.

4.3.1 Determination of moisture content of lead azide.-A tared dried aluminum dish, two inches in diameter shall be covered with a sheet of teflon (see specification MIL-P-14077), 2.5 inches square and sufficiently flat to form a tight cover. An accurately weighed specimen of approximately two grains shall be transferred to the moisture dish while both the cover and dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95 ± 2°C. for two hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool at room temperature. The teflon cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{(A-B)}{A} \times 100$$

Where:

A - the weight of the specimen before drying.
B - the weight of the specimen after drying.

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4.3.2 Determination of moisture content of PETN.-Approximately 5 grams of PETN shall be placed in a tared glass weighing dish. The dish and its contents shall be weighed, dried at 100 ± 2°C, for one hour, cooled in a desiccator and weighed. The loss of weight shall be calculated to the percentage of moisture in the PETN.

4.3.3 Resistance between the pin and cup.-The resistance shall be measured between the pin and the cup. An assembly that does not meet the applicable requirements shall be classed defective.

4.3.4 Waterproofness and functioning.

4.3.4.1 Conditioning.-The detonators, shall be submerged in water for the time, the temperature and the depth specified. The detonators shall be subjected to the following test within one hour after removal from the water.

4.3.4.2 Time and duds.-The detonator shall be assembled into test equipment as specified in 4.2.4 and functioned. Observation shall be made for satisfactory functioning and duds.

4.3.4.3 Output and duds.-The detonator shall be assembled into test equipment as specified in 4.2.4 and functioned. Observation shall be made for satisfactory functioning and duds.

4.3.4.4 Duds.-The detonator shall be assembled into test equipment as specified in 4.2.4 and functioned. Observation shall be made for satisfactory functioning and duds.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The detonators shall be preserved and packaged in accordance with dwg. 8800784.

5.1.2 Intermediate packaging.-Packaging shall be in accordance with dwg. 8798698.

5.2 Packing

5.2.1 Level C.-The detonators shall be packed in accordance with dwg. 8998699.

5.3 Marking.-Marking shall be in accordance with dwgs. P-8800784, P-8798698 and P-8798699, as applicable.

5.3.1 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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6.2 Standard deviation.-The following formual or its equivalent shall be used to compute the sample standard deviation:

$$s = \sqrt{\frac{\leqslant(\bar{x}-x)^2}{n-1}}$$

Where:

s - sample standard deviation.
$\bar{x}$ - sample mean time.
x - individual observed time.
$\leqslant(\bar{x}-x)^2$ - the sum of the square of the sample mean time minus each individual observed time.
n - sample size.

6.3 Check tests for possible deterioration.-The following tests are designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The tests should be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under averse conditions for any period of time. If the specifiestion for the ammunition item into which the detonators are loaded does not require that the check tests be performed, the contracting officer should insert a provision for the performance of the check tests in the contract. Two hundred detonators shall be selected for these tests.

6.3.1 Function time and duds.

6.3.1.1 Procedure.-Forty (40) detonators shall be subjected to the test specified in 4.3.4.2. Observation shall be made for the functioning time and all duds that may occur will be recorded. Duds will be replaced by detonators from the 200 selected.

6.3.1.2 Rejection.-If the mean time ($\bar{x}$) plus 2.03 times the standard deviation (s) exceeds 10 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in paragraph 6.2 or equivalent. Duds shall not enter into the calculation for standard deviation.

6.3.2 Output and Duds.

6.3.2.1 Procedure.-Fifty (50) detonators shall be subjected to the test specified in 4.3.4.3. Observation shall be made for the requirement for output and the number of duds that may occur will be recorded. Duds will be replaced by detonators from the 200 selected.

6.3.2.2 Retest and rejection.-If two detonators fail to adequately perforate the lead disc, a retest shall be permitted using the balance of the detonators selected in 6.3. If during the retest one more detonator fails to adequately perforate the lead disc, the retest shall be stopped and the lot rejected. During the retest the number of duds that may occur will be recorded. If three of more detonators fail to adequately perforate the lead disc during the original test the lot will be rejected without retest.

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6.3.3 Duds.

6.3.3.1 Procedure.-If no retest is required in the output phase, the balance of the detonators selected in 6.3 shall be subjected to the test for duds as specified in 4.3.4.4. The number of duds that may occur will be recorded.

6.3.3.2 Retest and rejection.-If more than two but not more than six duds are found a retest will be permitted using double the number of detonators selected in 6.3. The lot shall be rejected during retest as soon as the total number of duds observed in 6.3.1, 6.3.2 and 6.3.3 is equal to seven. If in the original sample of 200, seven or more duds are observed the lot shall be rejected without retest.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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